

Presentation Guide

This guide is intended for presenting the E-Commerce Shopping Cart System to stakeholders, instructors, or peers. It provides a structured flow to demonstrate the core features of the system effectively.

1. Introduction (2 mins)

- **Objective:** Explain the purpose of the application (a robust CLI e-commerce cart system).
- **Tech Stack:** Highlight that it is built with pure Java 8+ without external dependencies.
- **Key Features:** Mention inventory management, cart operations, tax/coupon calculations, and CSV persistence.

2. Setup & Execution (1 min)

- Show the directory structure (`src/` , `data/` , `output/`).
- Demonstrate how to compile and run the application using the terminal:

```
javac -d bin src/*.java
java -cp bin Main
```

- Mention that data is initially loaded from `data/products.csv` and `data/coupons.csv` .

3. Feature Walkthrough (5-7 mins)

Walk through the interactive CLI menu step-by-step:

A. Browsing Products

- Choose Option `1` .
- Explain how the table dynamically displays products loaded directly from the CSV file.

B. Cart Operations

- Choose Option `2` to Add an item (e.g., ID: `1` or `P001` , Quantity: `2`).
- Try adding a quantity that exceeds stock to show error handling.

- Choose Option `3` to View Cart and show current subtotal and tax.
- Choose Option `4` to Update Quantity and Option `5` to Remove an item. Show the cart again to verify changes.

C. Coupons & Discounts

- Choose Option `6` to apply a coupon.
- Select a valid coupon and show the success message.
- Explain the logic: the system validates the coupon's expiry date and applies either a percentage or flat discount.

D. Checkout & Persistence

- Choose Option `7` to Checkout.
- Show the final receipt printed in the console.
- **Crucial Step:** Open `data/products.csv` to prove that the stock quantities have been permanently deducted.
- Show that the receipt is also saved to `output/sample_outputs.txt`.

4. Code Architecture Highlight (2 mins)

- Briefly show `Main.java` for the CLI loop.
- Show `Cart.java` for the core business logic.
- Show `FileHandler.java` for the CSV parsing and data persistence.

5. Q&A (End)

- Open the floor to any questions regarding the design decisions, such as why CSV was used for persistence or how the code handles incorrect inputs without crashing.